

CONCURRENCY

- `.task(id:)`: view-scoped work, cancels & restarts when the id changes
- `@MainActor`: default actor isolation for unannotated code
- `@concurrent`: leaves the main actor for a nonisolated async function
- `AsyncStream`: wraps delegate callbacks for `for await`

SWIFTDATA

- `@Model`: persistent, observable class; `@Attribute`, `@Relationship`
- `@Query + #Predicate`: type-checked, live-updating fetch
- `ModelContainer/ModelConfiguration`: App Group + CloudKit store
- `VersionedSchema + SchemaMigrationPlan/MigrationStage` migrations

PREFERENCES, FILES & DOCUMENTS

- `UserDefaults/@appStorage`: property-list store, App Group suite
- `@SceneStorage`: transient per-scene UI state
- `Sandbox`: Documents, Application Support, Caches, tmp; `containerURL`
- `ReadableDocument/WritableDocument` (27), `FileDocument`, `UIDocument`

NETWORKING: URLSESSION

- `async data(for:)`, `bytes(from:)` (`AsyncBytes`), upload, download
- `JSONDecoder`; `URLError` `retry/backoff`; cancellation
- `URLSessionConfiguration.background + sessionSendsLaunchEvents`
- `NWPathMonitor`: `isExpensive/isConstrained`, Low Data Mode

BACKGROUND EXECUTION

- `BGTaskScheduler`: `BGAppRefreshTask`, `BGProcessingTask`, `BGContinuedProcessingTask` (26)
- `.backgroundTask(.appRefresh(_:)) / .backgroundTask(.processingTask(_:))` (27)
- `BGTaskSchedulerPermittedIdentifiers` in the `Info.plist`
- App Intents' `LongRunningIntent` (27)

NOTIFICATIONS & APNS

- `UNNotificationCenter.requestAuthorization`; provisional authorization
- `UNNotificationCategory/UNNotificationAction`, `UNNotificationCenterDelegate`
- APNs: device token, token-based (p8) auth, `apns-push-type`
- aps payload: alert, sound, badge (JSON, max 4 KB)

LIVE ACTIVITIES

- `ActivityAttributes`: static attributes + dynamic `ContentState`
- `ActivityConfiguration`: Lock Screen + Dynamic Island views
- `Activity.request(attributes:content:pushType:)`
- Live Activity push, `pushToStartTokenUpdates`, `LiveActivityIntent`

WIDGETKIT

- Provider returns `placeholder/snapshot/Timeline` of entries + reload policy
- `WidgetCenter.reloadTimelines(ofKind:)`
- `Button(intent:)/Toggle(isOn:intent:)` run an App Intent, then WidgetKit reloads
- `ControlWidget/ControlWidgetButton/ControlWidgetToggle` for Control Center

APP INTENTS & SYSTEM INTEGRATION

- `AppIntent`, `AppEntity/EntityQuery`, `perform()`
- `AppShortcutsProvider/AppShortcut`: Siri & Shortcuts phrases, no setup
- Universal links, custom URL scheme, `NSUserActivity`, `onOpenURL`
- Core Spotlight: `IndexedEntity`, `indexAppEntities`

FOUNDATION MODELS

- `LanguageModelSession`: a conversation with a model, keeps a transcript
- `@Generable`: derives a schema for guided generation
- `Tool`: `Sendable` type with a name, description, `@Generable` arguments
- `SystemLanguageModel`: `streamResponse`; context window, guardrails

LOCATION, MAPS & PHOTOS

- `CLLocationWhenInUseUsageDescription`; `CLLocationUpdate.liveUpdates`
- `NSPhotoLibraryUsageDescription` (read), `NSPhotoLibraryAddUsageDescription` (save)
- `NSCameraUsageDescription` (+ `NSMicrophoneUsageDescription` for video)
- MapKit for SwiftUI: `Map`, `Marker`, `MKMapItem`

SECURITY & PRIVACY

- Keychain Services (`SecItemAdd/SecItemCopyMatching`), `kSecAttrAccessible`
- `LocalAuthentication/LAContext`: Face ID, Touch ID, Optic ID
- Sign in with Apple, passkeys (`ASAuthorizationPlatformPublicKeyCredentialProvider`)
- `PrivacyInfo.xcprivacy`, required reason API, App Tracking Transparency

STOREKIT 2

- `Product.products(for:)`; `Product.purchase(options:)`
- `Transaction.updates` yields `VerificationResult<Transaction>` (JWS)
- `.verified/.unverified`; `finish()`; `currentEntitlements`
- `StoreView/SubscriptionStoreView`; `StoreKit` configuration file for testing

SWIFT TESTING

- `@Test/@Suite`: implicit suite, or explicit for a display name + traits
- `#expect` records and continues; `try #require` throws and stops the test
- Traits: `Tag`, `enabled(if:)`, `timeLimit`, `serialized`, `customTestScoping`
- Parameterized tests with `arguments:`; `withKnownIssue`; `Test.cancel`

DEBUGGING & INSTRUMENTS

- LLDB: `v` (frame variable, try first), `p/po` (expression/--object-description), `bt`
- Breakpoints: symbolic, Swift error, exception, runtime issue; conditions + log actions
- language swift task tree; view debugger; memory graph, malloc stack logging
- `xctrace` templates: Time Profiler, Animation Hitches, System Trace, Leaks, App Launch

SIGNING

- Code-signing identity: certificate (Apple Development/Distribution) + private key
- Provisioning profile answers who/what/where/when/how (TN3125)
- App ID (team prefix + bundle ID); registered devices; entitlements allowlist
- `CODE_SIGN_STYLE` `automatic/manual`; `FieldNotes.entitlements`

CLI: XCODEBUILD, NOTARYTOOL, SIMCTL, DEVICECTL

- `xcodebuild build/test/archive/-exportArchive` with `ExportOptions.plist`
- `xcodebuild build-for-testing/test-without-building, -testPlan`
- `simctl create/boot/bootstatus/status_bar`; `devicectl` for physical devices
- `notarytool submit --wait` for notarization

TESTFLIGHT → APP REVIEW

- Archive (Release, Apple Distribution) → Distribute App upload → processing (validation, thinning, symbols)
- TestFlight internal group: no review; external group: beta review (light, first build only)
- App Review Guidelines; submission and phased release
- App Privacy details, Accessibility Nutrition Labels, age-rating questionnaire